Chen Sun

csun8109@usc.edu | github.com/ChenSun-YP | linkedin.com/in/chen-sun-79671b134 | chensun-yp.github.io

EDUCATION

University of Southern California — **Los Angeles, CA**
Ph.D. student, Biomedical Engineering (expected Apr 2028) — Sept 2023–Present
Advisor: Dr. Dong Song

Queen's University — **Ontario, Canada**
B.S. degree: Computer Science — Sept 2018-May 2022

SKILLS

Programming & Tools: Python, MATLAB, SQL, Go, PyTorch, OpenCV, ROS, COMSOL.
Research & Modeling: Time-series analysis, signal processing, spiking neural networks, nonlinear and nonstationary modeling, reinforcement learning, deep learning, electromagnetic simulation, Transformer, CNN, ResNet, MLP, SSM, JEPA, VAE, SSL, multimodal sensor fusion, and neural time-series decoding.
Computing: HPC clusters, Slurm, distributed computing.
Agentic Tools: Codex, Claude Code, Cursor, Grok.

ACADEMIC EXPERIENCE

Neural Modeling and Interface Laboratory, USC — **Los Angeles, CA**
Research Assistant — Sept 2023-Present

- Led DARPA INSPIRE research identifying synaptic plasticity from in vivo recordings during learning; presented at SfN 2025.
    - Built PyTorch model to directly infer population-level LTP/LTD from hippocampal time-series recorded during cognitive learning tasks.
    - Processed large public electrophysiology datasets with neuron-pair recordings; trained models with mixed precision and multi-GPU PyTorch DistributedDataParallel (DDP) on Slurm-managed HPC clusters.
- Developed a graphon generative model using a Multiwavelet Neural Operator Transformer (MWNOT) for parameter inference across graph sizes and EEG functional-connectivity analysis.
- Worked with functional ultrasound imaging (fUSI) recordings from a ketamine-induced seizure animal model.
- Led electromagnetic simulation of a research project on how miniaturized TMS interacts with brain tissue and affects cognition in behaving rats.
    - Built COMSOL finite-element models from CT/MRI-derived rat anatomy, assigned tissue-specific electromagnetic properties, and quantified field strength, focality, and penetration across different head and brain volumes.
    - Published in Journal of Neural Engineering (2026).

Department of Biomedical Engineering, USC — **Los Angeles, CA**
Teaching Assistant — Sept 2023-Present

- BME 501-Intro to Biomedical Engineering, BME 513-Signal and System Analysis, BME 552-Neural Implant Engineering

Machine Intelligence & Biocomputing Laboratory (MIB), Queen's University — **Ontario, Canada**
Research Assistant — March 2022-April 2023

- Investigated the application of various evolutionary algorithms for pruning deep spiking neural networks (DSNNs).
- Researched the use of coevolution algorithms for optimizing DSNNs with multi-objective models.
- Examined the integration of dendrites into SNN models and developed a continuous learning capable dendritic DSNN model.

PUBLICATIONS AND MANUSCRIPTS

Google Scholar

Sun, C., Ren, B., Robinson, B. S., Moore, B., Zhang, X., & Song, D. Identification of Long-Term Synaptic Plasticity from Endogenous Spikes During Learning. *Manuscript in preparation.*

Jiang, W., Choi, W., Hsu, C., Lau, I., McHugh, M., **Sun, C.**, et al. (2026). Chronic subthreshold rTMS transiently disrupts learning but enhances memory and flexibility in healthy rats. *Journal of Neural Engineering, 23(5), 056035.*

Choi, W., Pinto, I. I. L. D., **Sun, C.**, Gupta, G., Song, D., & Bogdan, P. (2026). A Generative Model of Complex Networks Using Graphons and Neural Inverse Operators. arXiv:2610.02439 *(preprint).*

PROFESSIONAL EXPERIENCE

ByteDance **Beijing, China**

Full-Stack Developer Intern May 2021-Sept 2021

Developed backend systems for two internal platforms:

- Form-Filling Platform: Designed with Python's Django framework and Gunicorn for concurrency, enabling secure data collection via Lark and online Excel integration, supporting over 100 active users.
- HR Management Platform (HR+): Built a Golang/Hertz admin module for internal operations and an internal transfer platform, deployed company-wide across all HR departments.

Mesoor AI **Shanghai, China**

Web Crawler Developer Intern Aug 2020-Feb 2021

- Designed and developed a modular recruitment recommendation system to automate resume collection and job matching.
- Improved BERT resume analysis and targeted market-data crawlers to boost platform accuracy and efficiency.
- Built a post-promotion service and secure proxy generators for scalable data requests and user engagement.

SELECTED PROJECTS

LitAgent: Agentic Literature Intelligence Platform **Los Angeles, CA**

Personal Project Apr 2026-Present

- Built LitAgent, an end-to-end RAG-based literature assistant that ingests Zotero libraries, indexes papers, and delivers grounded QA with citations using user-provided LLMs as agents. Developed interactive wiki and relation-map interfaces with retrieval, graph construction, and orchestration modules for cross-paper synthesis.

GazeReader: Apple Vision Pro PDF Reader App **Los Angeles, CA**

Personal Project Nov 2025-Dec 2025

- Developed GazeReader, an immersive AR/VR application on visionOS using Swift, RealityKit, and ARKit.
- Implemented spatial PDF viewer with real-time highlighting of read text sections via eye-tracking and hand gestures.

EEG Foundation Challenge **Los Angeles, CA**

Deep Decoding Model for Cross-Task Cross-Subject EEG Decoding Sept 2025-Dec 2025

- Built a PyTorch model for EEG-based behavioral prediction, with subject- and session-specific embeddings to capture individual and session variability.
- Generated session-specific channel masks by mapping session embeddings through a two-layer MLP.
- Applied the masks to aggregate subject-specific EEG channels into a composite neural signal for downstream behavioral prediction.
- Ranked in the top 10% of participating teams.

Massachusetts Institute of Technology **Cambridge, MA**

Thalamic Model for Task Representation and Segmentation in Dynamic Environments June 2023-Present

- Built a PyTorch thalamic model for dynamic task segmentation with self-generated internal task representations.
- Trained in a custom OpenAI Gym environment with volatility and task switching to study task-boundary inference.
- Showed single unsegmented network could learn opposing tasks, achieve lifelong learning, and adapt to novel tasks without catastrophic forgetting.
- Predicted task-segmentation triggers in animal behavior; compared synaptic plasticity with neural-activity updates.

Queen's University **Ontario, Canada**

EvoMedley, Interactive Melody Generator with Multi-song Fusion Jan 2022-April 2022

- Proposed EvoMedley, an interactive music mixing method using evolutionary algorithms (EA) to generate personalized music from user-selected playlists.
- Developed MIDI feature extraction into minimal melodic fragments for an EA-based music-generation gene pool.
- Designed dynamic user-guided fitness for real-time preference adaptation while retaining original track features.

Tangent **Ontario, Canada**

Customized Online Archive May 2022-Sept 2022

- Led Tangent Collective's online archive: recruitment, project management, backend development, and deployment.
- Built a Django/MySQL backend with NGINX/WSGI on AWS EC2, S3 storage, CloudFront CDN, and RDS for data recoverability; used pm2 for stable automated deployment and daemonization.